

1.1 Random Binary Expansions

1 Approximating F

Question 1.

Fix $x \in [0, 1]$. We observe that $Y_j^n = \mathbb{1}[X_j^n < x]$ is a Bernoulli random variable with $\mathbb{P}[Y_j^n = 1] = F^{(n)}(x)$, where $F^{(n)}(x)$ is the cumulative distribution function of X^n (which should approach that of X as $n \rightarrow \infty$). Then $\sum_{j=1}^N Y_j^n \sim \text{Bin}(N, F^{(n)}(x))$, and using the standard formula for the variance of a binomial distribution,

$$\begin{aligned} \text{Var}(\hat{F}(x)) &= \frac{1}{N^2} \text{Var}\left(\sum_{j=1}^N Y_j^n\right) \\ &= \frac{1}{N} F^{(n)}(x)(1 - F^{(n)}(x)) \\ &\leq \frac{1}{4N} \end{aligned}$$

We choose N such that the standard deviation in $\hat{F}(x)$ at every point $x \in (0, 1)$ is at most 0.0005, which gives us $N = 10^6$.

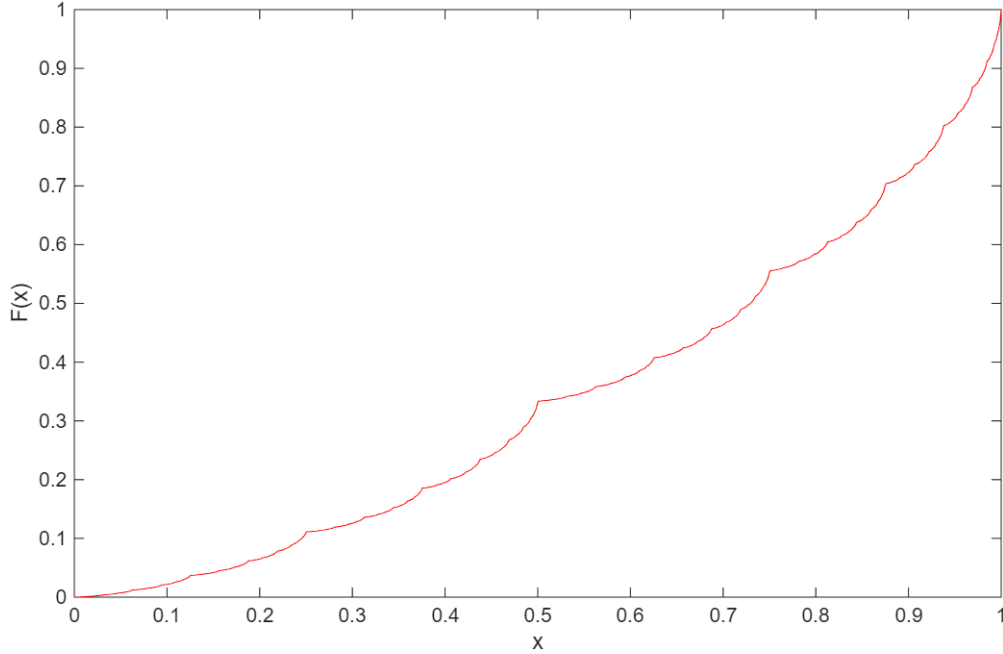


Figure 1: Empirical distribution function $\hat{F}(x)$ for $p = \frac{2}{3}$, with $n = 30$ and $N = 10^6$.

2 Calculating F

Question 2.

Fix $x = \sum_{i=1}^n \frac{x_i}{2^i} \in [0, 1]$ for some $n \in \mathbb{N}$. Let $X = \sum_{i=1}^{\infty} \frac{U_i}{2^i} \in [0, 1]$.

We see that $X < x$ iff there exists some $k \in \{0, 1, 2, \dots, n-1\}$ such that $U_i = x_i$ for all $i \leq k$ and $U_{k+1} < x_{k+1}$. Observe that, by this definition, k must be unique, so we have

$$\mathbb{P}[X < x] = \sum_{k=0}^{n-1} \mathbb{P}[\{U_{k+1} < x_{k+1}\} \cap \{U_i = x_i \ \forall i \leq k\}]$$

Now, $U_i \sim \text{Ber}(p)$ for all $i \in \mathbb{N}$, so

$$\begin{aligned} \mathbb{P}[U_i = x_i] &= \begin{cases} q & x_i = 0 \\ p & x_i = 1 \end{cases} \\ &= q^{1-x_i} p^{x_i} \\ \mathbb{P}[U_i < x_i] &= \begin{cases} 0 & x_i = 0 \\ q & x_i = 1 \end{cases} \\ &= qx_i \end{aligned}$$

Then, for some fixed k , since all the U_i are independent,

$$\begin{aligned} \mathbb{P}[\{U_{k+1} < x_{k+1}\} \cap \{U_i = x_i \ \forall i \leq k\}] &= \mathbb{P}[U_{k+1} < x_{k+1}] \prod_{i=1}^k \mathbb{P}[U_i = x_i] \\ &= qx_{k+1} \prod_{i=1}^k q^{1-x_i} p^{x_i} \end{aligned}$$

Thus

$$\mathbb{P}[X < x] = q \sum_{k=0}^{n-1} x_{k+1} \prod_{i=1}^k q^{1-x_i} p^{x_i}$$

Also, observe that $X = x$ only if X has a finite binary expansion of length n , which is true only if $U_i = 0$ for all $i > n$ or $U_i = 1$ for all $i > n$. Since $p, q \in (0, 1)$, this occurs with probability 0.

Then

$$\begin{aligned} F(x) &= \mathbb{P}[X \leq x] \\ &= \mathbb{P}[X < x] + \mathbb{P}[X = x] \\ &= \mathbb{P}[X < x] \end{aligned}$$

Therefore we have

$$F(x) = q \sum_{k=0}^{n-1} x_{k+1} \prod_{i=1}^k q^{1-x_i} p^{x_i} \tag{1}$$

Question 3.

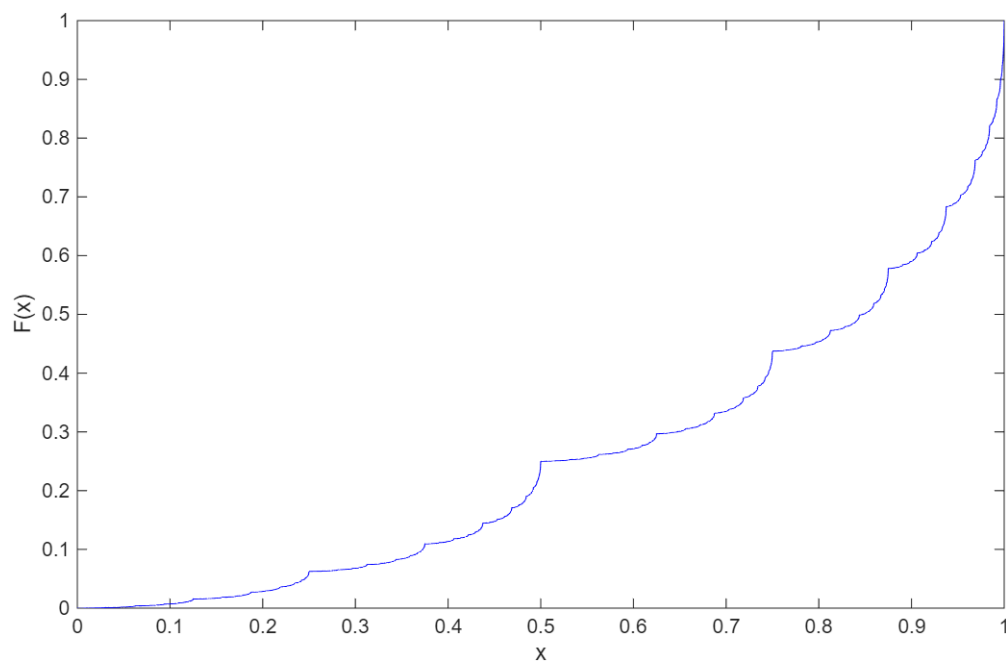


Figure 2: $F(x)$ for $p = \frac{3}{4}$ with $n = 11$ and $N = 10^6$.

We can overlay this on the graph obtained from question 1.

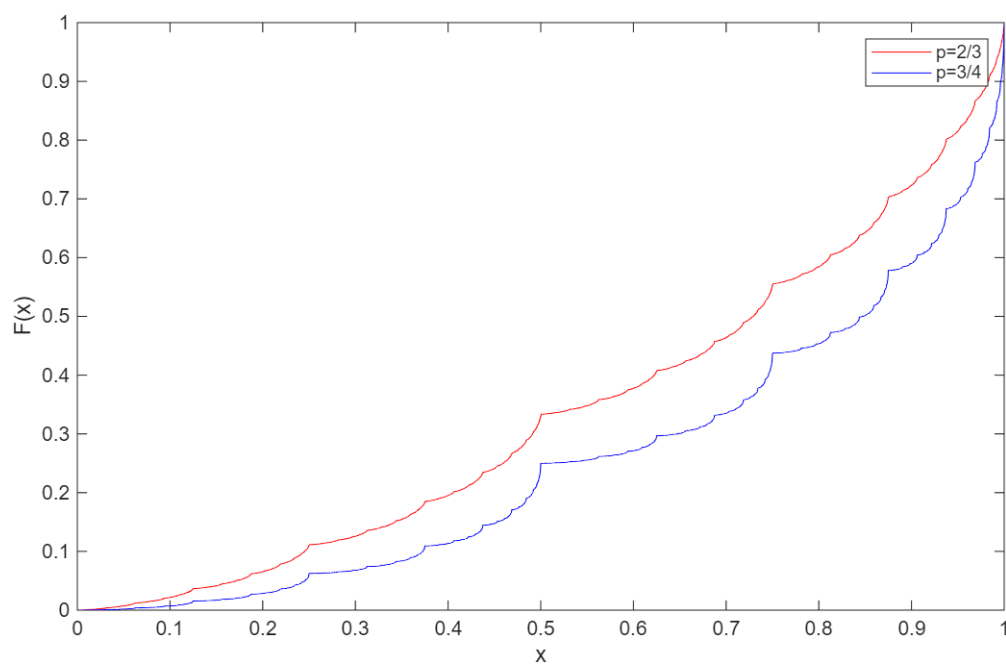


Figure 3: $F(x)$ for $p = \frac{3}{4}$ (in blue) and $p = \frac{2}{3}$ (in red)

We see that $F_{3/4}(x) \leq F_{2/3}(x)$ for all $x \in [0, 1]$, with equality at the points $x \in \{0, 1\}$. Also, the two graphs appear to have discontinuous “slope” at the same points. This is most obvious at $x = \frac{1}{2}$, and at $x = \frac{1}{4}, \frac{3}{4}$. This appears to be the case at any $x \in [0, 1]$ with a finite binary expansion (which we formally prove later in Question (6)), with a greater discontinuity at points with fewer digits in their binary expansion.

More generally, we can prove that if $p_1 \geq p_2$, then $F_{p_1}(x) \leq F_{p_2}(x)$ for any $x \in [0, 1]$, which explains our first observation above. To see this, we let $Z_i \sim U[0, 1]$ be uniformly distributed on $[0, 1]$. Define

$$\begin{aligned}\bar{U}_i^{(p_j)} &= \mathbb{1}[Z_i \leq p_j] \\ \bar{X}^{(p_j)} &= \sum_{i=1}^{\infty} \frac{\bar{U}_i^{(p_j)}}{2^i}\end{aligned}$$

Observe that $\bar{U}_i^{(p_j)}$ is a Bernoulli random variable with probability p_j of being equal to 1. Then $\bar{X}^{(p_j)}$ has the same distribution as the random variable X (with $p = p_j$), so we must have

$$F_{p_j}(x) = \mathbb{P}[\bar{X}^{(p_j)} < x]$$

Now, observe that $p_1 > p_2 \implies \bar{U}_i^{(p_1)} \geq \bar{U}_i^{(p_2)}$ for all $i \in \mathbb{N} \implies \bar{X}^{(p_1)} \geq \bar{X}^{(p_2)}$. Then for any $x \in [0, 1]$,

$$\{\bar{X}^{(p_1)} \leq x\} \subseteq \{\bar{X}^{(p_2)} \leq x\}$$

Thus $\mathbb{P}[\bar{X}^{(p_1)} < x] \leq \mathbb{P}[\bar{X}^{(p_2)} < x]$ and therefore $F_{p_1}(x) \leq F_{p_2}(x)$, as required.

Now we consider the complexity of evaluating F at a fixed point $x \in [0, 1]$ which has a finite binary expansion of length n . The method used in Question (1) involves generating N sample binary sequences of length n , converting the binary sequences into decimal values via the equation $X^n = \sum_{i=1}^n \frac{U_i}{2^i}$ (which involves $O(n)$ operations) and comparing the result with the value of x , for a total complexity of $O(nN)$.

The second algorithm evaluates the expression $q \sum_{k=0}^{n-1} x_{k+1} \prod_{i=1}^k q^{1-x_i} p^{x_i}$, which involves $\sum_{k=0}^{n-1} k \approx \frac{n^2}{2}$ multiplications, so the total complexity of the algorithm is $O(n^2)$.

3 Properties of F

Question 4.

We fix $p \in (0, 1)$. Let $q = 1 - p$ and $\gamma = \max\{p, q\}$. Then $\gamma < 1$.

F is continuous at a point $x \in [0, 1]$ if, for all $\epsilon > 0$, $\delta > 0$ such that

$$|F(y) - F(x)| < \epsilon \quad \forall |y - x| < \delta$$

We observe that F is a cumulative distribution function, so it is monotonically increasing. Thus, it suffices to prove that there exists $l < x < L$ such that

$$\begin{aligned}F(x) - F(l) &< \epsilon \\ F(L) - F(x) &< \epsilon\end{aligned}\tag{2}$$

Then continuity follows by taking $\delta = \max\{x - l, L - x\}$ and observing that

$$y \in [l, L] \implies F(y) \in [F(l), F(L)]$$

Let $c \in (0, 1)$ have a finite binary expansion and write

$$c = \sum_{i=1}^m \frac{c_i}{2^i}$$

Since $c \neq 0$, we can choose m such that $c_m = 1$. For $M > m$, we define

$$\begin{aligned} l^{(M)} &= c - \frac{1}{2^M} = \sum_{i=1}^M \frac{l_i^{(M)}}{2^i} \\ L^{(M)} &= c + \frac{1}{2^M} = \sum_{i=1}^M \frac{L_i^{(M)}}{2^i} \end{aligned} \tag{3}$$

where

$$\begin{aligned} l_i^{(M)} &= \begin{cases} c_i & 1 \leq i \leq m-1 \\ 0 & i = m \\ 1 & m+1 \leq i \leq M \end{cases} \\ L_i^{(M)} &= \begin{cases} c_i & 1 \leq i \leq m \\ 0 & m+1 \leq i \leq M-1 \\ 1 & i = M \end{cases} \end{aligned}$$

Substituting these expressions into our equation (1) for $F(x)$, we see that

$$\begin{aligned} F(c) - F(l^{(M)}) &= q \sum_{k=0}^{m-1} c_{k+1} \prod_{i=1}^k q^{1-c_i} p^{c_i} - q \sum_{k=0}^{M-1} l_{k+1}^{(M)} \prod_{i=1}^k q^{1-l_i^{(M)}} p^{l_i^{(M)}} \\ &= q \prod_{i=1}^{m-1} q^{1-c_i} p^{c_i} - q \sum_{k=m}^{M-1} l_{k+1}^{(M)} \prod_{i=1}^k q^{1-l_i^{(M)}} p^{l_i^{(M)}} \\ &= \left[q \prod_{i=1}^{m-1} q^{1-c_i} p^{c_i} \right] \left[1 - \sum_{k=m}^{M-1} l_{k+1}^{(M)} \prod_{i=m}^k q^{1-l_i^{(M)}} p^{l_i^{(M)}} \right] \\ &= \left[q \prod_{i=1}^{m-1} q^{1-c_i} p^{c_i} \right] \left[1 - \sum_{k=m}^{M-1} qp^{k-m} \right] \\ &= \left[q \prod_{i=1}^{m-1} q^{1-c_i} p^{c_i} \right] [p^{M-m}] \\ &\leq \gamma^M \rightarrow 0 \text{ as } M \rightarrow \infty \\ F(L^{(M)}) - F(c) &= q \sum_{k=0}^{M-1} L_{k+1}^{(M)} \prod_{i=1}^k q^{1-L_i^{(M)}} p^{L_i^{(M)}} - q \sum_{k=0}^{m-1} c_{k+1} \prod_{i=1}^k q^{1-c_i} p^{c_i} \\ &= q \sum_{k=m}^{M-1} L_{k+1}^{(M)} \prod_{i=1}^k q^{1-L_i^{(M)}} p^{L_i^{(M)}} \\ &= q \prod_{i=1}^{M-1} q^{1-L_i^{(M)}} p^{L_i^{(M)}} \\ &\leq \gamma^M \rightarrow 0 \text{ as } M \rightarrow \infty \end{aligned} \tag{4}$$

Then we see that for any $\epsilon > 0$, we can pick M such that

$$\begin{aligned} F(c) - F(l^{(M)}) &< \epsilon \\ F(L^{(M)}) - F(c) &< \epsilon \end{aligned}$$

and $l^{(M)} < c < L^{(M)}$, thus $F(x)$ is continuous at $x = c$.

For $c = 0$ and $c = 1$, observe that taking the same choice of $L^{(M)} = \frac{1}{2^M}$ (for $c = 0$) and $l^{(M)} = 1 - \frac{1}{2^M}$ (for $c = 1$) and following the argument above shows the continuity at the endpoints.

From our plots in figure (3), F appears to be continuous at all points in $[0, 1]$. Let $z \in [0, 1]$ have no finite binary expansion. Then we can write

$$z = \sum_{i=1}^{\infty} \frac{z_i}{2^i}$$

where we observe that $z_i = 0$ and $z_i = 1$ infinitely often (i.e. for all $N \in \mathbb{N}$, there exist $r, s > N$ such that $z_r = 0$ and $z_s = 1$). We define M_n to be the n^{th} integer i such that $z_i = 0$. Then $M_n \geq n$ for all $n \in \mathbb{N}$.

From the continuity condition given by (2), we observe that it is sufficient to find $l < z < L$ such that

$$F(L) - F(l) < \epsilon$$

Then condition (2) (and therefore continuity) follows from the observation that $F(x)$ is monotonically increasing, so $F(z) \in [F(l), F(L)]$. Now we define

$$l^{(n)} = \sum_{i=1}^{M_n-1} \frac{z_i^{(n)}}{2^i}$$

$$L^{(n)} = \frac{1}{2^{M_n}} + \sum_{i=1}^{M_n-1} \frac{z_i^{(n)}}{2^i}$$

Since $\exists i > M_n$ such that $z_i = 1$, we see that $l^{(n)} < z$. Also, since $z_{M_n} = 0$, we see that $L^{(n)} > z$. Now, both $l^{(n)}$ and $L^{(n)}$ have finite binary expansions, so substituting these expressions into equation (1), we see that

$$\begin{aligned} F(L) - F(l) &= q \sum_{k=0}^{M_n-1} L_{k+1}^{(n)} \prod_{i=1}^k q^{1-L_i^{(n)}} p^{L_i^{(n)}} - q \sum_{k=0}^{M_n-1} l_{k+1}^{(n)} \prod_{i=1}^k q^{1-l_i^{(n)}} p^{l_i^{(n)}} \\ &= q \prod_{i=1}^{M_n-1} q^{1-z_i} p^{z_i} \\ &\leq \gamma^{M_n} \\ &\leq \gamma^n \rightarrow 0 \text{ as } M \rightarrow \infty \end{aligned}$$

So for any $\epsilon > 0$ can pick n such that

$$F(L^{(n)}) - F(l^{(n)}) < \epsilon$$

and $l^{(n)} < z < L^{(n)}$, thus $F(x)$ is continuous at $x = z$, as expected.

Question 5.

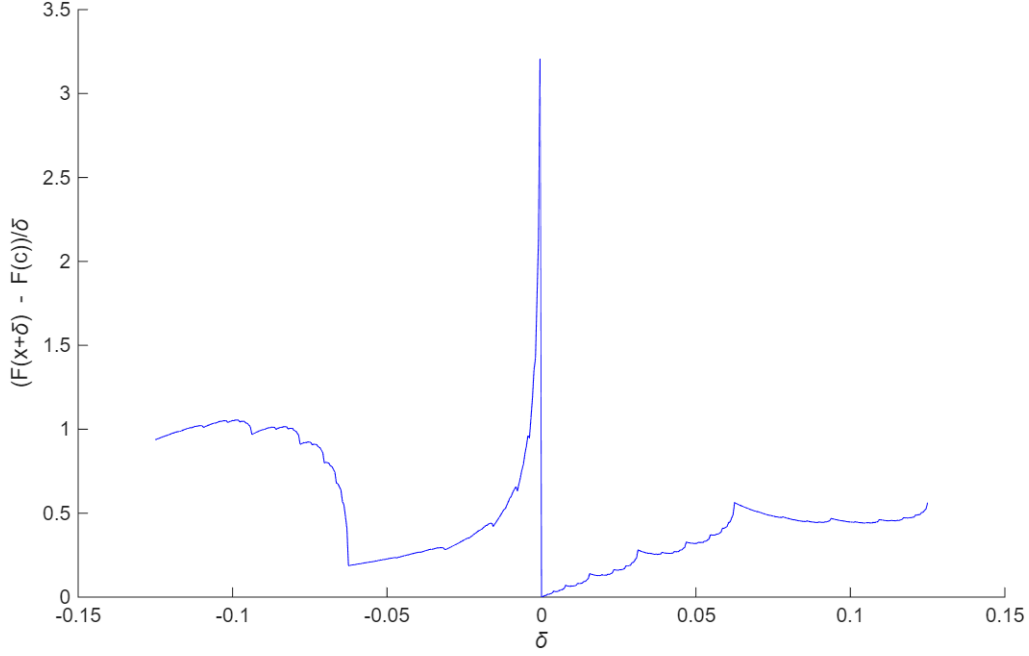


Figure 4: $\frac{F(c+\delta)-F(c)}{\delta}$ against δ for $\delta = \pm\frac{1}{2^{11}}, \dots, \pm\frac{256}{2^{11}}$,

Since the decimal expansion of $c = \frac{9}{16}$ terminates at $n = 4$, we choose δ to lie in the range $[-\frac{1}{2^3}, \frac{1}{2^3}]$. We observe that for small values of $|\delta|$, $\frac{F(c+\delta)-F(c)}{\delta}$ appears to diverge if $\delta < 0$, so we choose $\delta_{min} = \frac{1}{2^{11}}$, so that the behaviour of $\frac{F(c+\delta)-F(c)}{\delta}$ elsewhere can still be clearly seen on the plot.

Our plot seems to suggest that F is right-differentiable at $c = \frac{9}{16}$, with right-derivative 0, but not left-differentiable, as $\frac{F(c+\delta)-F(c)}{\delta}$ appears to diverge as $\delta \uparrow 0$.

Question 6.

First, we note that since $F(x)$ is monotonically increasing, we must have that

$$\frac{F(c+\delta) - F(c)}{\delta} \geq 0$$

for all $c \in [0, 1]$, $\delta \neq 0$.

We conjecture that, at an arbitrary point $c \in [0, 1]$ with finite binary expansion,

1. If $p > \frac{1}{2}$, F is right-differentiable at c but not left-differentiable.
2. If $p < \frac{1}{2}$, F is left-differentiable at c but not right-differentiable.
3. If $p = \frac{1}{2}$, F is differentiable at c .

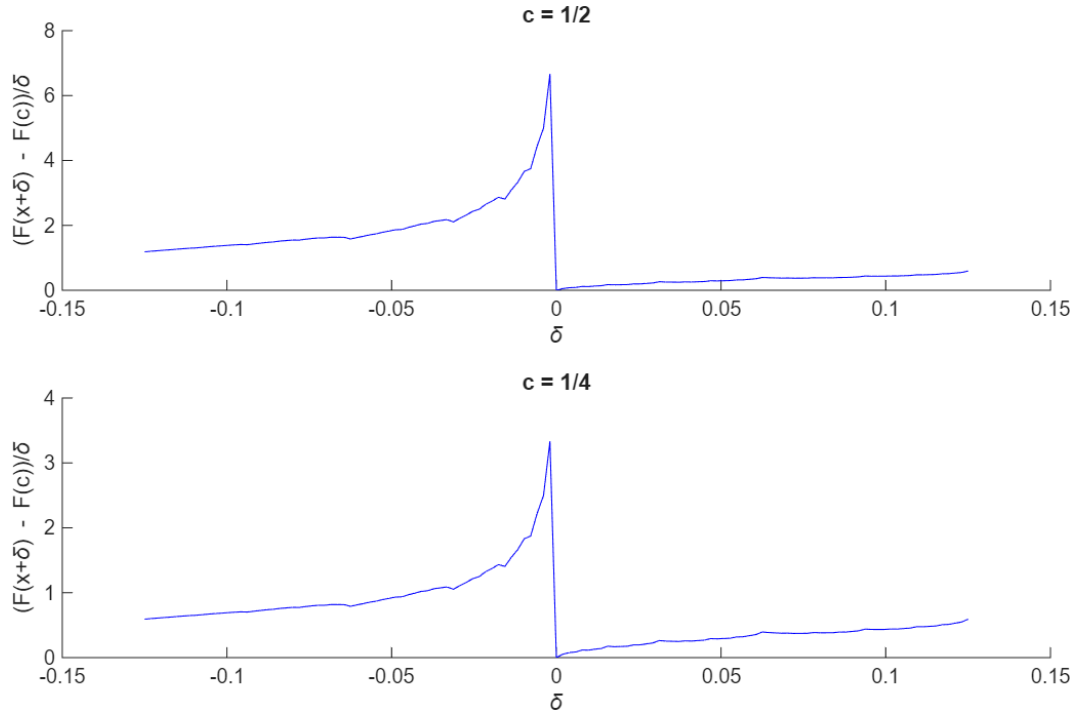


Figure 5: $\frac{F(c+\delta)-F(c)}{\delta}$ against δ for $p = \frac{2}{3}$ and $c = \frac{1}{4}, \frac{1}{2}$

As seen in Question (5), we see that if $p > \frac{1}{2}$, $\frac{F(c+\delta)-F(c)}{\delta}$ appears to tend to 0 as $\delta \downarrow 0$ but diverges as $\delta \uparrow 0$, suggesting that F is right-differentiable but not left-differentiable.

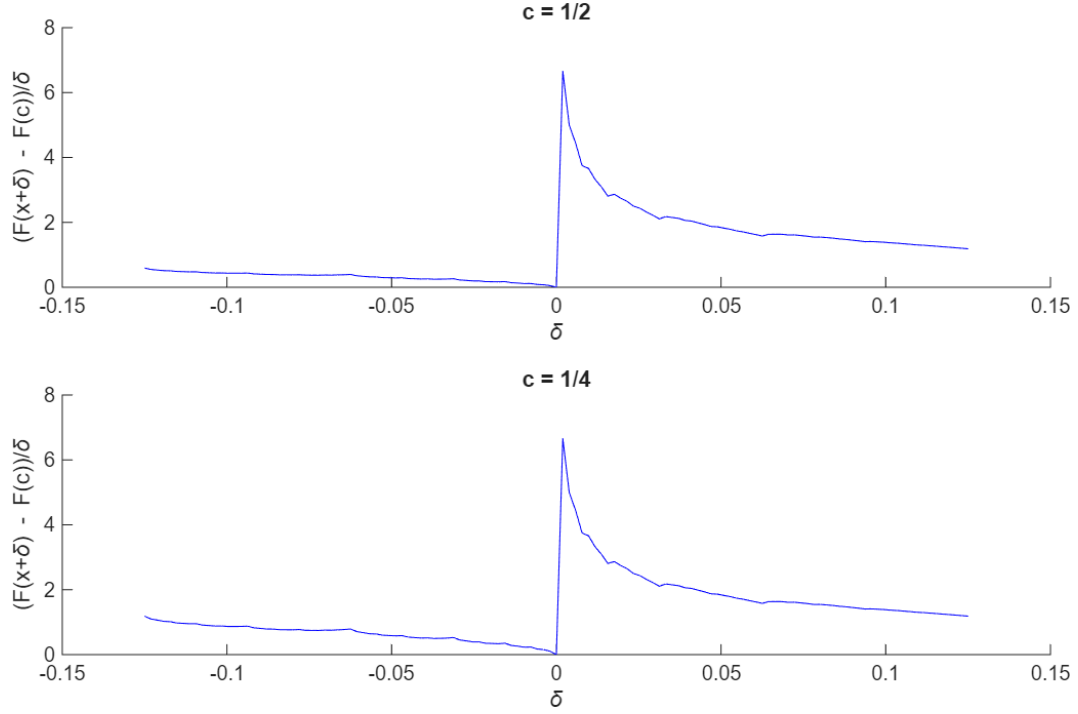


Figure 6: $\frac{F(c+\delta)-F(c)}{\delta}$ against δ for $p = \frac{1}{3}$ and $c = \frac{1}{4}, \frac{1}{2}$

For $p < \frac{1}{2}$, we appear to observe the opposite behaviour; $\frac{F(c+\delta)-F(c)}{\delta}$ appears to diverge as $\delta \downarrow 0$ but tends to 0 as $\delta \uparrow 0$, suggesting that F is left-differentiable but not right-differentiable.

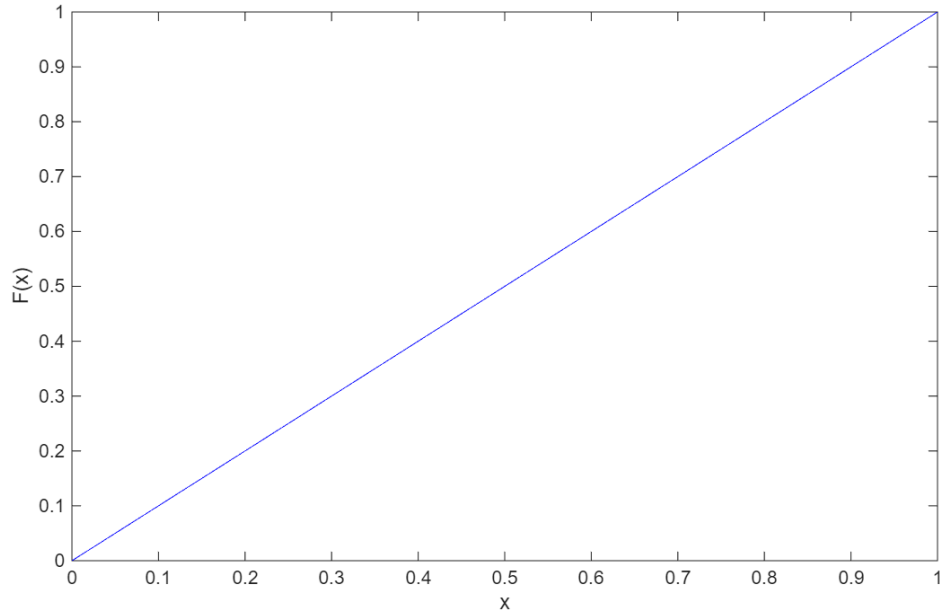


Figure 7: $F(x)$ against x for $p = \frac{1}{2}$

If $p = \frac{1}{2}$, we see from figure (7) that we appear to have $F(c) = c$ for all $c \in [0, 1]$ with finite binary

expansion, leading us to believe that $F(x)$ is everywhere differentiable on $[0, 1]$.

We now prove our conjecture. First, we look at the case $p > \frac{1}{2}$. Note that this implies $p > q = 1 - p$. Let

$$c = \sum_{i=1}^m \frac{c_i}{2^i}$$

and define for $M > m$

$$l^{(M)} = c - \frac{1}{2^M} = \sum_{i=1}^M \frac{l_i^{(M)}}{2^i}$$

$$L^{(M)} = c + \frac{1}{2^M} = \sum_{i=1}^M \frac{L_i^{(M)}}{2^i}$$

as before.

For $\delta > 0$ sufficiently small, we observe that there exists $M \in \mathbb{N}$ such that $\delta \in [\frac{1}{2^{M+1}}, \frac{1}{2^M}]$. Since $F(x)$ is monotonically increasing, we therefore have

$$\frac{F(c + \delta) - F(c)}{\delta} \leq \frac{F(L^{(M)}) - F(c)}{L^{(M+1)} - c} = 2 \frac{F(L^{(M)}) - F(c)}{L^{(M)} - c}$$

Then since $L^{(M)} \downarrow c$ as $M \rightarrow \infty$, it suffices to show that

$$\frac{F(L^{(M)}) - F(c)}{L^{(M)} - c} \rightarrow 0 \text{ as } M \rightarrow \infty$$

Referring to Equations (3) and (4) in Question (4), we see that

$$\begin{aligned} \frac{F(L^{(M)}) - F(c)}{L^{(M)} - c} &= 2^M q \prod_{i=1}^{M-1} q^{1-L_i^{(M)}} p^{L_i^{(M)}} \\ &= 2^M q \left[\prod_{i=1}^m q^{1-c_i} p^{c_i} \right] \left[\prod_{i=m+1}^{M-1} q \right] \\ &\leq 2^M [p^m] [q^{M-m}] \\ &= \frac{p^m}{q^m} (2q)^M \rightarrow 0 \text{ as } M \rightarrow \infty \end{aligned}$$

where the limit as $M \rightarrow \infty$ is 0 since $p > \frac{1}{2} \implies 2q < 1$. Thus we see that

$$\lim_{\delta \downarrow 0} \frac{F(c + \delta) - F(c)}{\delta} = 0$$

so the right-derivative at $x = c$ is 0.

For $\delta < 0$ sufficiently small, again there exists $M \in \mathbb{N}$ such that $\delta \in [-\frac{1}{2^{M-1}}, -\frac{1}{2^M}]$. Since $F(x)$ is monotonically increasing, we have that

$$\frac{F(c) - F(c + \delta)}{-\delta} \geq \frac{F(c) - F(l^{(M)})}{c - l^{(M-1)}} = \frac{1}{2} \frac{F(c) - F(l^{(M)})}{c - l^{(M)}}$$

Then since $l^{(M)} \uparrow c$ as $M \rightarrow \infty$, it suffices to show that

$$\frac{F(c) - F(l^{(M)})}{c - l^{(M)}} \rightarrow +\infty \text{ as } M \rightarrow \infty$$

Referring again to Equations (3) and (4) in Question (4), we see that

$$\begin{aligned}\frac{F(c) - F(l^{(M)})}{c - l^{(M)}} &= 2^M \left[q \prod_{i=1}^{m-1} q^{1-c_i} p^{c_i} \right] [p^{M-m}] \\ &\geq 2^M q^m p^{M-m} \\ &= \frac{q^m}{p^m} (2p)^M \rightarrow +\infty \text{ as } M \rightarrow \infty\end{aligned}$$

where the expression diverges as $M \rightarrow \infty$ since $2p > 1$. Thus we see that

$$\frac{F(c + \delta) - F(c)}{\delta} \rightarrow +\infty \text{ as } \delta \uparrow 0$$

and so the left-derivative does not exist, as hypothesised.

Next, we look at the case $p < \frac{1}{2}$. Following the same arguments as above (and noting that we now have $p < q = 1 - p$), we observe that that we have

$$\begin{aligned}\delta > 0 \implies \frac{F(c + \delta) - F(c)}{\delta} &\geq \frac{F(L^{(M)}) - F(c)}{L^{(M-1)} - c} \text{ for some } M \text{ with } \delta \in \left[\frac{1}{2^M}, \frac{1}{2^{M-1}}\right] \\ &= \frac{1}{2} \frac{F(L^{(M)}) - F(c)}{L^{(M)} - c} \\ &= \frac{2^M}{2} q \left[\prod_{i=1}^m q^{1-c_i} p^{c_i} \right] \left[\prod_{i=m+1}^{M-1} q \right] \\ &\geq \frac{2^M}{2} p^m q^{M-m} \\ &= \frac{p^m}{2q^m} (2q)^M \rightarrow \infty \text{ as } M \rightarrow \infty \\ \delta < 0 \implies \frac{F(c + \delta) - F(c)}{\delta} &\leq \frac{F(c) - F(l^{(M)})}{c - l^{(M+1)}} \text{ for some } M \text{ with } \delta \in \left[-\frac{1}{2^M}, -\frac{1}{2^{M+1}}\right] \\ &= 2 \frac{F(c) - F(l^{(M)})}{c - l^{(M)}} \\ &= 2^{M+1} \left[q \prod_{i=1}^{m-1} q^{1-c_i} p^{c_i} \right] [p^{M-m}] \\ &\leq 2^{M+1} q^m p^{M-m} \\ &= \frac{2q^m}{p^m} (2p)^M \rightarrow 0 \text{ as } M \rightarrow \infty\end{aligned}$$

where the first expression diverges and the second expression tends to 0 since $2q > 1$ and $2p < 1$. Thus we see that $F(x)$ is left-differentiable with left-derivative 0, but not right-differentiable, as hypothesised.

Finally, we look at the case $p = \frac{1}{2}$. As before, we write

$$c = \sum_{i=1}^m \frac{c_i}{2^i}$$

Substituting this into equaiton (1) with $p = q = \frac{1}{2}$, we get

$$\begin{aligned} F(c) &= \frac{1}{2} \sum_{k=0}^{m-1} c_{k+1} \prod_{i=1}^k \frac{1}{2} \\ &= \sum_{k=0}^{m-1} \frac{c_{k+1}}{2^{k+1}} \\ &= c \end{aligned}$$

as expected.

We claim that this implies $F(x) = x$ for all $x \in [0, 1]$. Let $z \in [0, 1]$ have no finite binary expansion. We write

$$z = \sum_{i=1}^{\infty} \frac{z_i}{2^i}$$

We define $l^{(n)}, L^{(n)}$ as in Question (4). Letting M_n be the n^{th} integer i such that $z_i = 0$,

$$\begin{aligned} l^{(n)} &= \sum_{i=1}^{M_n-1} \frac{z_i^{(n)}}{2^i} \\ L^{(n)} &= \frac{1}{2^{M_n}} + \sum_{i=1}^{M_n-1} \frac{z_i^{(n)}}{2^i} \end{aligned}$$

Since $l^{(n)}, L^{(n)}$ have finite binary expansions, we have $F(l^{(n)}) = l^{(n)}$ and $F(L^{(n)}) = L^{(n)}$. Observe that $l^{(n)} < z < L^{(n)}$, with $z - l^{(n)} < \frac{1}{2^{M_n}}$ and $L^{(n)} - z < \frac{1}{2^{M_n}}$. Then we must have

$$\begin{aligned} F(z) &> F(l^{(n)}) = l^{(n)} > z - \frac{1}{2^{M_n}} \\ F(z) &< F(L^{(n)}) = L^{(n)} < z + \frac{1}{2^{M_n}} \end{aligned}$$

So $|F(z) - z| < \frac{1}{2^{M_n}} \leq \frac{1}{2^n}$. But since this is true for all $n \in \mathbb{N}$, we must have $|F(z) - z| = 0$, and our claim that $F(x) = x$ for all $x \in [0, 1]$ holds. Then $F(x)$ is differentiable on $[0, 1]$ with derivative 1, as hypothesised.

4 Program Listings

Question 1: Empirical $\hat{F}$

```
N = 10 .^ (6); %Size of random sample
M = 0; %Loop counter

n = 30; %Length of binary expansion

p = 2./3;
q = 1-p; %Parameters

plotpoints = 1000; %Number of points to plot
xplot = linspace(0, 1, plotpoints);
yplot = zeros(1, plotpoints);

while M < N
    m = 1; %Loop counter
    %Generate X_j
    ytemp = 0;
    while m <= n
        flip = rand;
        if flip < p
            ytemp = ytemp + 2 .^ (-m);
        end
        m = m+1;
    end

    i = 0; %Loop counter
    %Increment all points with x>X_j
    while i < plotpoints
        i = i+1;
        if xplot(i) >= ytemp
            yplot(i) = yplot(i) + 1./N;
        end
    end

    M = M+1;
end

plot(xplot,yplot, 'r')
xlabel('x')
ylabel('F(x)')
```

Question 3: Analytic F

```
n = 11; %Length of binary expansion
p = 3./4;
q = 1-p; %Parameters
l = zeros(1,n+1); %Index (vector of binary digits)
xplot = linspace(0,1, 2.^n+1);
yplot = zeros(1,2.^n + 1);
yplot(2.^n + 1) = 1;

while l(1) < 1
    s = 0; %Loop counter
    %Convert index (vector of binary digits) to decimal integer
    for r = 1:n+1
        s = s + 2.^(n+1-r) .* l(r);
    end

    %Calculate F
    for k = 0:n-1
        %No need to evaluate the product term if x_(k+1) is 0
        if l(k+2) == 0 %Our indexing is one off (we included the coeff of 2^0)
            continue;
        end
        prodtemp = 1; %Temporary variable
        for i = 1:k
            if l(i+1) == 0
                prodtemp = prodtemp * q; %Update product term
            else
                prodtemp = prodtemp * p; %Update product term
            end
        end
        yplot(s+1) = yplot(s+1) + q.*prodtemp;
    end

    %Increment index by one
    for j = 0:n
        l(n+1-j) = mod(l(n+1-j)+1,2);
        if l(n+1-j) == 1
            break;
        end
    end
end

plot(xplot,yplot, 'b')
xlabel('x')
ylabel('F(x)')
```

Question 5: $(F(c + \delta) - F(c))/\delta$

```

n = 11; %Length of binary expansion
m = 3; %Maximum value of delta is 2^-m
p = 3./4;
q = 1-p; %Parameters
c = zeros(1,n+1);
c(2) = 1;
c(5) = 1; %Binary expansion for c
Fc = q + p .* (q.^3); %Value of F(c) (hardcoded for c=9/16)

l = zeros(1,n+1); %Index (vector of binary digits)

xplot = zeros(1,2.^(n-m)+1); %For delta > 0
yplot = zeros(1,2.^(n-m) + 1);
xplot(1) = 0;
yplot(1) = 0;
nxplot = zeros(1,2.^(n-m)+1); %For delta<0
nyplot = zeros(1,2.^(n-m) + 1);
nxplot(1) = 0;
nyplot(1) = 0;

while l(m+1) < 1

    %Increment index by one
    for j = 0:n
        l(n+1-j) = mod(l(n+1-j)+1,2);
        if l(n+1-j) == 1
            break;
        end
    end

    s = 0;
    %Convert index to integer
    for r = 1:n+1
        s = s + 2.^(n+1-r) .* l(r);
    end

    %Find corresponding value of delta (decimal number)
    d = s ./ (2.^n);
    xplot(s+1) = d;
    nxplot(s+1) = -d;

    %Addition of c and delta (as vectors of binary digits)
    x = c+l;
    for t = 0:n
        if x(n+1-t) == 2
            x(n+1-t) = 0;
            x(n-t) = x(n-t)+1;
        end
        if x(n+1-t) == 3
            x(n+1-t) = 1;
            x(n-t) = x(n-t)+1;
        end
    end

    %Evaluate F(x)
    for k = 0:n-1
        %No need to evaluate the product term if x_(k+1) is 0
        if x(k+2) == 0 %Our indexing is one off because we in
            continue;
        end
        prodtemp = 1;
        for i = 1:k
            if x(i+1) == 0
                prodtemp = prodtemp * q; % Update product ter
            else
                prodtemp = prodtemp * p; % Update product ter
            end
        end
    end
end

```

```

%Subtraction of c and delta (as vectors of binary digits)
x = c-1;
for t = 0:n
    if x(n+1-t) == -1
        x(n+1-t) = 1;
        x(n-t) = x(n-t)-1;
    end
    if x(n+1-t) == -2
        x(n+1-t) = 0;
        x(n-t) = x(n-t) - 1;
    end
end

%Evaluate F(x)
for k = 0:n-1
    %No need to evaluate the product term if x_(k+1) is 0
    if x(k+2) == 0 %Our indexing is one off because we included the coeff of 2^0
        continue;
    end
    prodtemp = 1;
    for i = 1:k
        if x(i+1) == 0
            prodtemp = prodtemp * q; % Update product term with probability p
        else
            prodtemp = prodtemp * p; % Update product term with probability q
        end
    end
    nyplot(s+1) = nyplot(s+1) + q.*prodtemp;
end

%(F(x)-F(c)) / d
nyplot(s+1) = nyplot(s+1) - Fc;
nyplot(s+1) = -nyplot(s+1) ./ d;
end

hold on;
plot(xplot,yplot, 'b')
plot(nxplot,nyplot,'b')
xlabel('\delta')
ylabel('(F(x+\delta) - F(c))/\delta')
ax.XAxisLocation = 'origin';
ax.YAxisLocation = 'origin';
hold off;

```